LECTURE 02

Membership Table Examples

Examples 1–15

Example 1 — Union

$$A = \{1, 2, 4\}, \quad B = \{2, 3, 4\}$$

Find $A \cup B$ using a membership table.

Element	A	B	$A \cup B$
1	1	0	1
2	1	1	1
3	0	1	1
4	1	1	1

$$A \cup B = \{1, 2, 3, 4\}$$

Example 2 — Intersection

$$A = \{1, 2, 3, 5\}, \quad B = \{2, 3, 4, 6\}$$

Element	A	B	$A \cap B$
1	1	0	0
2	1	1	1
3	1	1	1
4	0	1	0
5	1	0	0
6	0	1	0

$$A \cap B = \{2, 3\}$$

Example 3 — Complement

$$U = \{1, 2, 3, 4, 5, 6\}, \quad A = \{1, 3, 5\}$$

Element	A	A^c
1	1	0
2	0	1
3	1	0
4	0	1
5	1	0
6	0	1

$$A^c = \{2, 4, 6\}$$

Example 4 — Difference

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}$$

Element	A	B	$A - B$
1	1	0	1
2	1	0	1
3	1	1	0
4	1	1	0
5	0	1	0
6	0	1	0

$$A - B = \{1, 2\}$$

Example 5 — Difference Identity

Use a membership table to show:

$$A - B = A \cap B^c$$

A	B	$A - B$	B^c	$A \cap B^c$
1	1	0	0	0
1	0	1	1	1
0	1	0	0	0
0	0	0	1	0

The corresponding final columns are identical; therefore, the identity is verified.

Example 6 — Symmetric Difference

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}, \quad A \cap B = \{3, 4\}$$

The common elements 3 and 4 belong to both sets, so they are excluded. The remaining elements belong to exactly one of the two sets.

$$A \triangle B = \{1, 2, 5, 6\}$$

Equivalent form.

$$A \triangle B = (A - B) \cup (B - A)$$
$$A - B = \{1, 2\}, \quad B - A = \{5, 6\}$$
$$A \triangle B = \{1, 2\} \cup \{5, 6\} = \{1, 2, 5, 6\}$$

> **Simple rule:** Symmetric difference = elements in A or B, but **not** in both.

Use a membership table to show $A \triangle B$:

A	B	$A \triangle B$
1	1	0
1	0	1
0	1	1
0	0	0

An element is in $A \triangle B$ if it belongs to exactly one of the two sets.

Example 7 — De Morgan's Law
Use a membership table to show:

$$(A \cup B)^c = A^c \cap B^c$$

A	B	$A \cup B$	$(A \cup B)^c$	A^c	B^c	$A^c \cap B^c$
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	0	0
0	0	0	1	1	1	1

The corresponding final columns are identical; therefore, the identity is verified.

Example 8 — Second De Morgan's Law
Use a membership table to show:

$$(A \cap B)^c = A^c \cup B^c$$

A	B	$A \cap B$	$(A \cap B)^c$	A^c	B^c	$A^c \cup B^c$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

The corresponding final columns are identical; therefore, the identity is verified.

Example 9 — Distributive Law

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

The two sides have identical membership values for every case.

Example 10 — Absorption Law

$$A \cup (A \cap B) = A$$

A	B	$A \cap B$	$A \cup (A \cap B)$	A
1	1	1	1	1
1	0	0	1	1
0	1	0	0	0
0	0	0	0	0

Example 11 — Another Absorption Law

$$A \cap (A \cup B) = A$$

A	B	$A \cup B$	$A \cap (A \cup B)$	A
1	1	1	1	1
1	0	1	1	1
0	1	1	0	0
0	0	0	0	0

Example 12 — Difference of Sets

$$A - (B \cup C) = (A - B) \cap (A - C)$$

A	B	C	$A-(B \cup C)$	$A-B$	$A-C$	$(A-B) \cap (A-C)$
1	1	1	0	0	0	0
1	1	0	0	0	1	0
1	0	1	0	1	0	0
1	0	0	1	1	1	1
0	1	1	0	0	0	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

Example 13 — Difference and Intersection

$$A - (B \cap C) = (A - B) \cup (A - C)$$

A	B	C	$A-(B \cap C)$	$A-B$	$A-C$	$(A-B) \cup (A-C)$
1	1	1	0	0	0	0
1	1	0	1	0	1	1
1	0	1	1	1	0	1
1	0	0	1	1	1	1
0	1	1	0	0	0	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

Example 14 — Associative Law of Union

$$(A \cup B) \cup C = A \cup (B \cup C)$$

A	B	C	$A \cup B$	$(A \cup B) \cup C$	$B \cup C$	$A \cup (B \cup C)$
1	1	1	1	1	1	1
1	1	0	1	1	1	1
1	0	1	1	1	1	1
1	0	0	1	1	0	1
0	1	1	1	1	1	1
0	1	0	1	1	1	1
0	0	1	0	1	1	1
0	0	0	0	0	0	0

Example 15 — Associative Law of Intersection

$$(A \cap B) \cap C = A \cap (B \cap C)$$

A	B	C	$A \cap B$	$(A \cap B) \cap C$	$B \cap C$	$A \cap (B \cap C)$
1	1	1	1	1	1	1
1	1	0	1	0	0	0
1	0	1	0	0	0	0
1	0	0	0	0	0	0
0	1	1	0	0	1	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

Rule. To prove two sets are equal using a membership table, construct the columns for both sides. If the final two columns are identical, the two sets are equal.